

Math 6601 Homework 9

Jacob Hauck

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Problem 1.

Let $f \in C^1([0, 1])$. Then

$$\left| \int_0^1 f(x) \, dx - \sum_{i=0}^{N-1} f(x_i)h \right| \leq \frac{Mh}{2}, \quad (1)$$

where $h = \frac{1}{N}$, and $x_i = ih$, and $M = \max_{0 \leq x \leq 1} |f'(x)|$.

Proof. We have

$$\int_0^1 f(x) \, dx = \sum_{i=0}^{N-1} \int_{x_i}^{x_{i+1}} f(x) \, dx,$$

and

$$f(x_i)h = \int_{x_i}^{x_{i+1}} f(x_i) \, dx.$$

Hence,

$$\begin{aligned} \left| \int_0^1 f(x) \, dx - \sum_{i=0}^{N-1} f(x_i)h \right| &= \left| \sum_{i=0}^{N-1} \int_{x_i}^{x_{i+1}} (f(x) - f(x_i)) \, dx \right| \\ &\leq \sum_{i=0}^{N-1} \int_{x_i}^{x_{i+1}} |f(x) - f(x_i)| \, dx. \end{aligned}$$

Let i be given and let $x \in [x_i, x_{i+1}]$. By the mean value theorem there exists c such that

$$f(x) - f(x_i) = f'(c)(x - x_i).$$

Then $|f(x) - f(x_i)| = |f'(c)|(x - x_i) \leq M(x - x_i)$ for any i and $x \in [x_i, x_{i+1}]$. Thus,

$$\begin{aligned} \left| \int_0^1 f(x) \, dx - \sum_{i=0}^{N-1} f(x_i)h \right| &\leq M \sum_{i=0}^{N-1} \int_{x_i}^{x_{i+1}} (x - x_i) \, dx \\ &\leq M \sum_{i=0}^{N-1} \frac{1}{2} (x_{i+1} - x_i)^2 \\ &\leq \frac{MNh^2}{2} = \frac{Mh}{2}. \end{aligned}$$

□

Problem 2.

Let $Q_m(f)$ be an m -point Gaussian quadrature rule on the compact interval $[a, b]$ with continuous weight function ρ . If f is continuous, then $Q_m(f) \rightarrow J(f)$ as $m \rightarrow \infty$, where

$$J(f) = \int_a^b f(x)\rho(x) \, dx.$$

Proof. Let $\varepsilon > 0$ be given. By the Weierstrass approximation theorem there exists a polynomial $p(x)$ such that $|f(x) - p(x)| < \frac{\varepsilon}{2\|\rho\|_1}$ for $x \in [a, b]$, where

$$\|\rho\|_1 = \int_a^b |\rho(x)| \, dx.$$

Let M be the degree of p . Then Q_m is exact for p for any $m > M$; that is, $Q_m(p) = J(p)$ if $m > M$. Then if $m > M$,

$$|Q_m(f) - J(f)| = |Q_m(f - p) + Q_m(p) - J(f - p) - J(p)| = |Q_m(f - p) - J(f - p)|$$

because Q_m and J are linear. Then

$$\begin{aligned} |Q_m(f) - J(f)| &= |Q_m(f - p) + J(f - p)| \leq |Q_m(f - p)| + |J(f - p)| \\ &\leq \sum_{i=1}^m |\alpha_i| |f(x_i) - p(x_i)| + \int_a^b |f(x) - p(x)| |\rho(x)| \, dx, \end{aligned}$$

where $\alpha_1, \dots, \alpha_m$ are the weights and $x_1, \dots, x_m$ are the nodes of the quadrature rule Q_m . By Theorem 6.3 in the textbook, we have $\alpha_i > 0$ for all i . Moreover,

$$\sum_{i=1}^m \alpha_i = Q_m(1) = J(1) = \|\rho\|_1$$

by the exactness of the Q_m for polynomials of degree $0 \leq M < m$. Thus,

$$\begin{aligned} |Q_m(f) - J(f)| &\leq \sum_{i=1}^m \alpha_i \frac{\varepsilon}{2\|\rho\|_1} + \int_a^b \frac{\varepsilon}{2\|\rho\|_1} |\rho(x)| \, dx \\ &\leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \end{aligned}$$

if $m > M$. This implies that $Q_m(f) \rightarrow J(f)$ as $m \rightarrow \infty$. □

Problem 3.

(a) For the composite trapezoid rule, we have

$$\int_0^{0.8} f(x) \, dx \approx \frac{h}{2} [f(0.0) + 2f(0.2) + 2f(0.4) + 2f(0.6) + f(0.8)],$$

where $h = \frac{0.8}{4} = 0.2$. Then we get

$$\int_0^{0.8} f(x) \, dx \approx 0.1 [1 + 2e^{-0.2} + 2e^{-0.4} + 2e^{-0.6} + e^{-0.8}] \approx 0.552505.$$

(b) For composite Simpson's rule, we have

$$\int_0^{0.8} f(x) \, dx \approx \frac{2h}{6} [f(0.0) + 4f(0.2) + 2f(0.4) + 4f(0.6) + f(0.8)],$$

where $h = \frac{0.8}{4} = 0.2$. Then we get

$$\int_0^{0.8} f(x) \, dx \approx \frac{0.2}{3} [1 + 4e^{-0.2} + 2e^{-0.4} + 4e^{-0.6} + e^{-0.8}] \approx 0.550676.$$

(c) Since $f(x) = e^{-x}$, the true value of the integral is

$$\int_0^{0.8} f(x) \, dx = -e^{-x} \Big|_0^{0.8} = 1 - e^{-0.8}.$$

Computing the error for the trapezoid rule, we have

$$E_T(f; h) \approx 1.834\,348 \times 10^{-3}.$$

Computing the error for Simpson's rule, we have

$$E_S(f; h) \approx 4.871\,642 \times 10^{-6}$$

Simpson's rule is considerably more accurate.

(d) Note that $f^{(2)}(x) = e^{-x}$, which has maximum absolute value 1 on $[0, 0.8]$ when $x = 0$. We want

$$E_T(f; h) \leq \frac{0.8 - 0}{12} \left(\frac{0.8 - 0}{n} \right)^2 < 10^{-4}.$$

Solving for n , we see that

$$n > \sqrt{\frac{(0.8)^3}{12} \cdot 10^4} \approx 20.66,$$

so $n + 1 = 22$ or more points is enough to guarantee an error of at most 10^{-4} using the trapezoid rule.

For Simpson's rule, we note that $f^{(4)}(x) = e^{-x}$, which also has maximum absolute value 1 on $[0, 0.8]$ when $x = 0$. We want

$$E_S(f; h) \leq \frac{0.8 - 0}{180} \left(\frac{0.8 - 0}{2n} \right)^4 < 10^{-4}.$$

Solving for n , we see that

$$n > \frac{1}{2} \left(\frac{(0.8)^5}{180} \cdot 10^4 \right)^{\frac{1}{4}} \approx 1.03,$$

so $2n + 1 = 5$ or more points is enough to guarantee an error of at most 10^{-4} using Simpson's rule.

Problem 4.

If the quadrature rule has degree of precision 3, then it must be exact for $p_i(x) = x^i$ for $i = 0, 1, 2, 3$. Then

$$\begin{aligned} a(1) + b(1) + c(1) &= a + b + c = \int_{-1}^1 1 \, dx = 2 \\ a(-1) + b(0) + c(1) &= -a + c = \int_{-1}^1 x \, dx = 0 \\ a(-1)^2 + b(0)^2 + c(1)^2 &= a + c = \int_{-1}^1 x^2 \, dx = \frac{2}{3} \\ a(-1)^3 + b(0) + c(1)^3 &= -a + c = \int_{-1}^1 x^3 \, dx = 0. \end{aligned}$$

The second and fourth equations are both equivalent to $a = c$, and the third equation gives $a = c = \frac{1}{3}$. Then by the first equation $b = 2 - \frac{2}{3} = \frac{4}{3}$.